

Evaluation of Tracking Systematic Uncertainties via Roadset Variance

Step-by-Step Methodology for the Drell-Yan Absolute Cross-Section

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Motivation: Tracking Systematics

Objective: Quantify the systematic uncertainty introduced by different trigger road definitions (RS57, RS59, RS62, RS67, RS70).

- **Challenge:** Different roadsets have drastically different Protons on Target (POT) and differing trigger acceptances.
- **Approach:** Calculate the single differential cross-section independently for each roadset, then evaluate the bin-by-bin variance.
- **Constraint:** Because the reliability of high-statistics roadsets (e.g., RS67) is under evaluation, we must treat roadsets equally to prevent biased pulls.

Step 1 & 2: Independent Execution & Dynamic Normalization

To prevent normalization distortions (the "Global POT Trap"), the analysis pipeline was refactored to execute *per roadset*.

Each execution dynamically overrides the global POT with the roadset-specific POT to yield the true independent cross-section:

Roadset	LH2 POT	LD2 POT	Flask POT
RS57	3.53×10^{16}	1.76×10^{16}	3.92×10^{15}
RS59	9.37×10^{15}	4.32×10^{15}	1.01×10^{15}
RS62	5.28×10^{16}	2.38×10^{16}	1.10×10^{16}
RS67	1.61×10^{17}	7.69×10^{16}	3.66×10^{16}
RS70	1.79×10^{16}	8.75×10^{15}	3.84×10^{15}

Step 3: Variance Calculation Mathematics

For a given p_T bin, let x_i be the absolute cross-section for roadset i , and e_i be its statistical error.

Inverse-Variance Weighting:

To suppress massive Poisson fluctuations from low-POT datasets (e.g., RS59), we scale by statistical power ($w_i = 1/e_i^2$):

$$\mu_w = \frac{\sum w_i x_i}{\sum w_i}$$

$$\sigma_{\text{sys},w} = \sqrt{\frac{\sum w_i}{(\sum w_i)^2 - \sum w_i^2} \sum w_i (x_i - \mu_w)^2}$$

Step 4: Resulting Uncertainties (LH2)

Using the inverse-variance weighted strategy suppresses low-statistics noise, yielding the following systematic errors for Liquid Hydrogen.

p_T	Bin	Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
	0	[0.00, 0.32)	0.001794	0.000290	16.18%
	1	[0.32, 0.49)	0.003872	0.000511	13.19%
	2	[0.49, 0.63)	0.004383	0.000536	12.24%
	3	[0.63, 0.77)	0.004766	0.000651	13.65%
	4	[0.77, 0.95)	0.004313	0.000533	12.37%
	5	[0.95, 1.18)	0.003721	0.000662	17.79%
	6	[1.18, 1.80)	0.001853	0.000248	13.40%

Step 4: Resulting Uncertainties (LD2)

Applying the same inverse-variance weighted procedure to Liquid Deuterium targets yields comparable systematic margins.

p_T	Bin	Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
	0	[0.00, 0.32)	0.003968	0.000698	17.59%
	1	[0.32, 0.49)	0.008777	0.001154	13.15%
	2	[0.49, 0.63)	0.010581	0.001340	12.67%
	3	[0.63, 0.77)	0.010224	0.001331	13.02%
	4	[0.77, 0.95)	0.010045	0.001534	15.27%
	5	[0.95, 1.18)	0.007935	0.001482	18.67%
	6	[1.18, 1.80)	0.004107	0.000738	17.98%

Why is the relative error still $\sim 13\text{--}18\%$?

- 1 **Statistical Noise Limitations:** While inverse-variance weighting heavily suppresses the noise of low-POT datasets (RS59, RS70), it cannot completely eliminate the pull from extreme outliers.
- 2 **Global Acceptance Mismatch:** A single `acceptance_mass_xF.root` file is currently used across all roadsets. Because different triggers have distinct kinematic acceptances, dividing by a global average creates artificial variance.

Path Forward: Implement roadset-specific Monte Carlo acceptance maps and evaluate the necessity of excluding extremely low-POT datasets from the overall variance calculation.